

Shape Optimization

Course notes: Sessions 1–11

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English teaching draft

Each session has a 90-minute course and a separate 90-minute lab. These notes reorganize the existing introductory chapters and the 2025 manuscripts. Sessions 1–2 introduce problems and examples; systematic existence, uniqueness and optimality arguments begin in Session 3. Sessions 5–6 develop the planar isoperimetric inequality and its equality case. Sessions 7–8 introduce convergence and compactness of shapes, then existence for convex shapes and the minimum-width problem. Session 9 develops mixed areas and volumes, Brunn–Minkowski and a new proof of the isoperimetric inequality. Sessions 10–11 study constant width and the Blaschke–Lebesgue theorem, with mixed-area, geometric and variational proofs. Prerequisites: elementary Euclidean geometry, single-variable calculus, vectors and basic linear algebra. Gradients are introduced here. Sessions 7–8 also use compactness, uniform convergence and elementary integration; Arzelà–Ascoli is stated in the form needed for the selection proof.

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1 Introduction: what is shape optimization?

By the end of the session: identify a design variable, an objective and an admissible class; formulate geometric optimization problems; distinguish an observed candidate from a verified optimum.

1.1 From numbers to geometric objects

The problem $\min_{x \in \mathbb{R}} (x + 1)^2$ asks us to choose a number. Its minimum value is 0, attained at the unique minimizer $x = -1$. In shape optimization the variable is a geometric object: a point, a curve, a polygon, or a domain $\Omega \subset \mathbb{R}^d$. We write

$$\inf_{\Omega \in \mathcal{A}} J(\Omega).$$

Here $\mathcal{A}$ is the *admissible class*, and J is the *objective* or *cost functional*. A functional assigns a number to each admissible object. We write a minimum when an admissible object actually attains the infimum. Maximizing J is equivalent to minimizing $-J$.

Design variable	Objective	Admissibility requirements
Point X	Distance $ X - Q $	X lies on a given line
Curve γ	Length $L(\gamma)$	Fixed endpoints
Planar domain Ω	Perimeter $P(\Omega)$	Area $ \Omega = A_0 > 0$
Polygon	Enclosed area	Fixed number of sides and perimeter
Material layout	Deflection under a load	Supports, load and amount of material fixed

Discussion. For each row, what changes if the constraint is removed? What measurements would be needed to evaluate the objective?

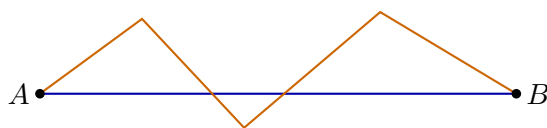
1.2 A first geometric solution: shortest paths

Let $A, B \in \mathbb{R}^2$. Among rectifiable continuous curves joining them, the segment $[A, B]$ has the least length, namely $|A - B|$. For a polygonal path this follows from the triangle inequality:

$$|A - P_1| + |P_1 - P_2| + \cdots + |P_k - B| \geq |A - B|.$$

The length of a continuous curve is the supremum of the lengths of its inscribed polygonal paths, so the same lower bound applies. The segment attains it. Curves with infinite length cannot improve this value.

This argument illustrates a useful strategy: find a bound valid for every competitor, then exhibit a competitor attaining it. We have proved global optimality, not merely checked a few possible paths.



Every detour has length at least $|A - B|$.

1.3 A catalogue of shape questions

Enclosing area. A loop of string has length L . Which closed curve encloses the largest area? Compare a triangle, a square and a circle. Session 2 explores the isoperimetric problem; a polygonal version returns in Lab 4.

A hanging chain. Fix two endpoints and the length of a flexible uniform chain. The chain seeks a low gravitational potential energy. Its shape depends on both the energy and the length constraint.

Networks. Given three towns A, B, C , choose a junction T to minimize $|T-A| + |T-B| + |T-C|$. Lab 2 studies the Torricelli point, also called the Fermat point.

Surfaces and structures. Surface area with a fixed enclosed volume models an idealized isolated bubble. A loaded structure suggests minimizing deformation with a fixed amount of material. These models can require differential equations to evaluate the objective; those equations are beyond the first four sessions.

1.4 How can a shape vary?

- *Parametric optimization:* change finitely many numbers, such as the side lengths of a rectangle or the vertices of a polygon.
- *Boundary variation:* deform the boundary of a domain while preserving its topology.
- *Topology variation:* allow components or holes to appear or disappear, subject to the chosen admissibility rules.

The first four sessions mostly use points, curves and finite-dimensional parameters. They introduce the questions that persist for more general shapes.

1.5 The questions to keep separate

1. *Existence:* is the best value attained?
2. *Uniqueness:* can more than one object attain it?
3. *Identification:* how can we find or characterize a minimizer?
4. *Approximation:* what does a computed candidate actually tell us?

Symmetries matter. If a disk solves an area-constrained perimeter problem, every translated disk solves it too. Uniqueness of a shape may mean uniqueness *up to rigid motions*.

Discussion. A program compares one hundred shapes and returns the best one. What has it established? Could a better shape be missing from its list?

Exit task. Formulate one shape optimization problem using $\mathcal{A}$ and J . State which quantities are fixed and which may vary.

2 Basic examples: good candidates and missing solutions

By the end of the session: state the two planar isoperimetric formulations, model a hanging chain, and recognize that a plausible optimal shape does not prove existence.

2.1 The isoperimetric problem

Let Ω be a bounded planar domain with a smooth simple closed boundary. Write $A = |\Omega|$ for its area and $P = P(\Omega)$ for its perimeter. The planar isoperimetric inequality is

$$P^2 \geq 4\pi A,$$

with equality precisely for disks. We state this result here; its proof is outside this introductory session. It answers two problems:

$$\min_{|\Omega|=A_0} P(\Omega) = 2\sqrt{\pi A_0}, \quad \max_{P(\Omega)=L} |\Omega| = \frac{L^2}{4\pi}.$$

The optimal radii are $\sqrt{A_0/\pi}$ and $L/(2\pi)$, respectively.

Scaling. Under a dilation by $t > 0$, area becomes $t^2 A$ and perimeter becomes tP . Thus P^2/A is unchanged. Rescaling competitors explains why the two formulations are equivalent.

For perimeter $L = 12$, an equilateral triangle has area $4\sqrt{3} \approx 6.93$, a square has area 9, and a disk has area $36/\pi \approx 11.46$. These comparisons support the disk as a candidate but do not compare it with every possible shape.

2.2 Heuristics and their logical limits

A visible inward dent suggests that replacing the dent by a straighter boundary can save length. One may then use scaling to restore a prescribed perimeter while increasing area. Symmetrization suggests that a best shape should be balanced in many directions.

The logical conclusion of an improvement argument is usually *if a maximizer exists, it must have certain properties*. This does not yet produce a maximizer. An infinite sequence of improvements might fail to converge, or might converge to an excluded object.

A straight shoreline. If only the free part of the boundary is charged, reflect a candidate across the shoreline. A free boundary of length L gives a doubled region of perimeter $2L$ and area $2A$. The stated isoperimetric inequality suggests the half-disk, with $A = L^2/(2\pi)$. This model permits the two shoreline endpoints to move. Fixing those endpoints introduces a further constraint.

2.3 The catenary: which quantity is minimized?

Consider an ideal perfectly flexible, inextensible chain of uniform mass per unit arc length in a uniform gravitational field, with no obstacles. Use an upward vertical coordinate y . For a graph $y : [a, b] \rightarrow \mathbb{R}$, the arc-length element is

$$ds = \sqrt{1 + y'(x)^2} dx.$$

After dropping a positive constant mass-density/gravity factor, its energy is

$$E[y] = \int_a^b y(x) \sqrt{1 + y'(x)^2} dx.$$

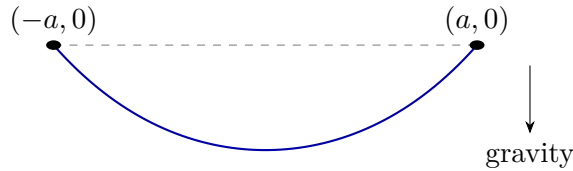
We seek to minimize this energy subject to fixed endpoint heights and

$$\int_a^b \sqrt{1 + y'(x)^2} dx = L.$$

The length must be at least the distance between the endpoints. For a slack chain with horizontally separated endpoints, the familiar equilibrium profile is a catenary

$$y(x) = c \cosh\left(\frac{x - x_0}{c}\right) + d, \quad c > 0.$$

The constants fit the endpoints and the length. We introduce the formula as the physical candidate; we do not derive an Euler–Lagrange equation or prove existence for this variational problem in this session.



For equal-height endpoints at $(-a, 0)$ and $(a, 0)$, symmetry gives $x_0 = 0$, $d = -c \cosh(a/c)$, and direct integration gives

$$L = 2c \sinh(a/c).$$

This is an equation for the remaining parameter, not an optimality proof. For $L = 2a$ the only shortest geometric path is the horizontal segment; the displayed slack catenaries approach it as $c \rightarrow \infty$.

Discussion. Why would minimizing chain length answer a different question? Why is $\int_a^b y(x) dx$ a different mass model?

2.4 Three ways a best value can fail to be attained

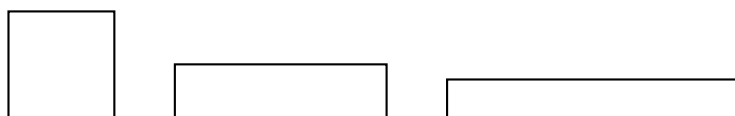
An excluded endpoint. On $(0, 1]$, $f(x) = x$ has infimum 0. The values at $x_k = 1/k$ approach 0, but no admissible point realizes it.

Escape to infinity. On $\mathbb{R}$, $f(x) = e^x$ has infimum 0, approached at $x_k = -k$. There is no finite limiting minimizer.

An unbounded objective. Rectangles with sides t and A_0/t all have area A_0 , while

$$P(t) = 2 \left(t + \frac{A_0}{t} \right) \rightarrow +\infty \quad (t \rightarrow +\infty).$$

Thus maximizing perimeter at fixed area has no finite optimum, even within convex polygons. A constraint can make a problem meaningful without guaranteeing that it has a solution.



Same area; increasing perimeter and diameter.

A warning about pictures of convergence. Draw a straight segment and then increasingly fine triangular teeth of amplitude proportional to their wavelength. The teeth approach the segment in position while their total length stays larger. A visually close boundary need not have a close perimeter. Session 3 will explain what kind of limit inequality is enough for minimization.



Straight length 1; toothed length $\sqrt{2}$.

For k equal teeth on a segment of length 1, choose tooth height $1/(2k)$. Each sloping half-tooth has length $\sqrt{2}/(2k)$, so the total length is $\sqrt{2}$ for every k , even though the heights tend to zero.

Exit task. Explain the difference between “every optimizer would be a disk” and “a disk attains the optimum.” Name one obstruction to the existence of a solution.

3 Optimization theory: existence, uniqueness and optimality

By the end of the session: use the minimizing-sequence argument, distinguish convexity from strict convexity, and apply first-order conditions in one and several variables.

3.1 Minimum values and minimizers

Let $K \subset \mathbb{R}^n$ be nonempty and $f : K \rightarrow \mathbb{R}$. A global minimizer x^* satisfies $f(x^*) \leq f(x)$ for every $x \in K$. A local minimizer satisfies this comparison only for feasible x in some neighborhood of x^* . The number $\inf_K f$ is the infimum value; it need not be attained and may equal $-\infty$.

If $m = \inf_K f$ is finite, a *minimizing sequence* is a sequence $(x_k) \subset K$ such that $f(x_k) \rightarrow m$. We can choose $m \leq f(x_k) < m + 1/k$ by the definition of infimum. The existence of this sequence alone does not give a minimizer.

3.2 The classical existence argument

Theorem 3.1 (Weierstrass). *If $K \subset \mathbb{R}^n$ is nonempty and compact, and $f : K \rightarrow \mathbb{R}$ is continuous, then f attains its minimum and maximum on K .*

Proof. First f is bounded below: otherwise choose $x_k \in K$ with $f(x_k) \leq -k$. Compactness gives a subsequence converging to $x^* \in K$; continuity would give $f(x_k) \rightarrow f(x^*)$ along that subsequence, a contradiction. Thus $m = \inf_K f$ is finite. Choose a minimizing sequence. Compactness again gives $x_{k_j} \rightarrow x^* \in K$, and continuity gives

$$f(x^*) = \lim_{j \rightarrow \infty} f(x_{k_j}) = m.$$

Apply the same argument to $-f$ for the maximum. □

In $\mathbb{R}^n$, compactness is equivalent to being closed and bounded. Boundedness gives a convergent subsequence; closedness keeps its limit in the feasible set. This equivalence must not be transferred without justification to spaces of curves or shapes.

The direct method in the calculus of variations. The proof has three structural steps:

1. Choose an admissible minimizing sequence for a finite infimum.
2. Extract a convergent subsequence whose limit is admissible.
3. Pass to the limit in the objective, or use a suitable one-sided inequality.

For the last step, continuity can be weakened to *lower semicontinuity*: if $x_k \rightarrow x$, then

$$f(x) \leq \liminf_{k \rightarrow \infty} f(x_k).$$

Indeed, $m \leq f(x^*) \leq \liminf_j f(x_{k_j}) = m$. For variational problems, both convergence and compactness must be chosen and proved in the relevant function or shape space. The finite-dimensional argument is a template, not an automatic proof for the catenary.

Why the inequality points this way. A downward jump in cost at the limit is harmless for minimization. An upward jump may destroy attainment. For example, on $[0, 1]$, set $f(0) = 1$ and $f(x) = x$ for $x > 0$. The feasible set is compact but the infimum 0 is not attained: lower semicontinuity fails at 0.

3.3 Coercivity replaces boundedness of the whole feasible set

A function is *coercive on K* if $f(x) \rightarrow +\infty$ whenever $x \in K$ and $\|x\| \rightarrow \infty$. If K is nonempty and closed and f is continuous and coercive on K , then a minimizer exists. Choose $x_0 \in K$; the sublevel set

$$K_0 = \{x \in K : f(x) \leq f(x_0)\}$$

is nonempty, closed and bounded. Minimize on K_0 by Weierstrass; points outside K_0 cannot improve the result.

Discussion. Which assumption fails for $f(x) = x$ on $(0, 1]$? For e^x on $\mathbb{R}$? Does a continuous function on a closed set always attain a minimum?

3.4 Convexity and uniqueness

A set K is *convex* if $(1-t)x + ty \in K$ for all $x, y \in K$ and $t \in [0, 1]$. A function on a convex set is *convex* if

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

It is *strictly convex* if the inequality is strict when $x \neq y$ and $0 < t < 1$.

Proposition 3.2. *On a convex feasible set, every local minimum of a convex function is global. A strictly convex function has at most one global minimizer.*

Proof. If x is a local minimizer and $f(y) < f(x)$, convexity gives $f((1-t)x + ty) < f(x)$ for every $0 < t < 1$, contradicting local minimality for small t . If distinct x, y both minimize a strictly convex function with value m , their midpoint has value strictly less than m , again a contradiction. $\square$

Convexity alone allows multiple minimizers: a constant function is an example. Strict convexity alone does not prove existence: e^x is strictly convex on $\mathbb{R}$ and has no minimizer. Convexity of the feasible set is essential: x^2 has two minimizers on $\{-1, 1\}$.

3.5 First-order conditions in one variable

Theorem 3.3 (Fermat's necessary condition). *If f is differentiable at an interior local minimizer x^* of an interval, then $f'(x^*) = 0$.*

Proof. For sufficiently small $h > 0$, the quotient $(f(x^* + h) - f(x^*))/h$ is nonnegative; for $h < 0$ it is nonpositive. Their common limit must be zero. $\square$

Stationarity is necessary here, but it is not sufficient: x^3 is stationary at 0 without having an extremum. Also, $-x^2$ is stationary at its maximum. At a boundary minimum the derivative need not vanish: $f(x) = x$ on $[0, 1]$ is minimized at 0. A nondifferentiable minimum, such as $|x|$ at 0, requires a different argument.

For a differentiable convex function on an interval, the supporting-line inequality

$$f(y) \geq f(x) + f'(x)(y - x)$$

shows that $f'(x) = 0$ is sufficient for global minimality. To obtain it, apply convexity at $x + t(y - x)$, divide by $t > 0$ and let $t \downarrow 0$. The same argument works with a gradient in $\mathbb{R}^n$.

3.6 Several variables and a quadratic example

For a differentiable f on an open subset of $\mathbb{R}^n$, collect the partial derivatives into

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right).$$

At an interior local minimizer, restricting f to each coordinate line and applying the one-dimensional result gives $\nabla f(x^*) = 0$. For a differentiable convex function, the inequality $f(y) \geq f(x) + \nabla f(x) \cdot (y - x)$ makes a stationary point a global minimizer on its convex domain.

For example, $f(x, y) = x^2 + y^2 + x + 1$ has gradient $(2x + 1, 2y)$. Its stationary point is $(-1/2, 0)$, and completing squares gives $f(x, y) = (x + 1/2)^2 + y^2 + 3/4$, proving uniqueness and global optimality.

More generally, let $Q \in \mathbb{R}^{n \times n}$ be symmetric positive definite and $b \in \mathbb{R}^n$. For

$$f(x) = \frac{1}{2}x^T Qx - b^T x$$

we have $\nabla f(x) = Qx - b$ and the unique minimizer $x^* = Q^{-1}b$. The identity

$$f(x) - f(x^*) = \frac{1}{2}(x - x^*)^T Q(x - x^*)$$

proves the claim directly. Alternatively, if $\alpha > 0$ is the least eigenvalue of Q , then $f(x) \geq (\alpha/2)\|x\|^2 - \|b\|\|x\|$ proves coercivity, and positive definiteness proves strict convexity.

Exit task. Give one example for each statement: no minimizer; multiple minimizers; a stationary point that is not a minimum.

4 Constraints, gradients and Lagrange multipliers

By the end of the session: interpret the gradient geometrically, use Taylor's formula, and form Lagrange systems with one or several regular equality constraints.

4.1 Why constraints change the optimality condition

Minimizing $x^2 + y^2 + z^2$ freely gives the origin. With $x + y + z = 1$, the origin is inadmissible. A constrained minimizer need not satisfy $\nabla f = 0$, because only motions along the feasible set are allowed. Fixed area, perimeter, endpoint positions and total material are natural constraints in shape optimization.

We study smooth equality constraints here. Positivity of side lengths still belongs in the feasible set; when the candidate lies in the open positive region, equality multipliers can be applied locally there. General inequality-constrained optimality conditions are a later topic.

4.2 Taylor's formula and steepest ascent

Differentiability at x means

$$f(x+h) = f(x) + \nabla f(x) \cdot h + o(\|h\|), \quad h \rightarrow 0.$$

For $f \in C^2$ near x , a second-order version is

$$f(x+h) = f(x) + \nabla f(x) \cdot h + \frac{1}{2}h^T D^2 f(x) h + o(\|h\|^2).$$

In a fixed unit direction d , the directional derivative is $D_d f(x) = \nabla f(x) \cdot d$. Cauchy-Schwarz gives

$$\nabla f(x) \cdot d \leq \|\nabla f(x)\|, \quad \|d\| = 1.$$

If the gradient is nonzero, equality occurs for $d = \nabla f(x)/\|\nabla f(x)\|$. Thus the gradient gives the direction of greatest first-order increase per unit Euclidean distance. Its negative gives the greatest decrease. At a stationary point all first-order directional derivatives vanish; higher-order behavior matters.

If $\nabla f(x) \cdot d < 0$, Taylor's formula implies $f(x+td) < f(x)$ for all sufficiently small $t > 0$. It does not justify an arbitrarily large step. For instance, for $f(x) = x^2$ at $x = 1$, the step $x - tf'(x) = 1 - 2t$ decreases f exactly when $0 < t < 1$.

4.3 Gradients are normal to regular level sets

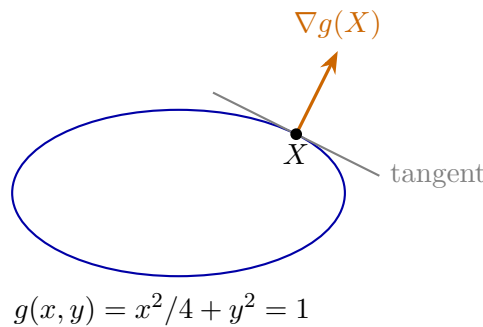
Suppose a differentiable curve γ stays on $g = c$ and $\gamma(0) = x$. Differentiating $g(\gamma(t)) = c$ gives

$$\nabla g(x) \cdot \gamma'(0) = 0.$$

At a regular point, meaning $\nabla g(x) \neq 0$, the tangent space of the level set is therefore

$$T_x\{g = c\} = \{d : \nabla g(x) \cdot d = 0\}.$$

That this kernel is exactly the tangent space uses the local regular level-set theorem. In the plane it is the tangent line.



A tangent vector preserves the constraint to first order. The straight step $x + td$ usually does not remain on a curved level set. Feasible motion follows a curve on the level set, with this vector as its initial velocity.

4.4 One equality constraint

Theorem 4.1 (Lagrange multiplier, one constraint). *Let f, g be C^1 on an open subset of $\mathbb{R}^n$. If x^* locally minimizes f subject to $g(x) = 0$, and $\nabla g(x^*) \neq 0$, then some $\lambda \in \mathbb{R}$ satisfies*

$$\nabla f(x^*) + \lambda \nabla g(x^*) = 0, \quad g(x^*) = 0.$$

Geometric explanation. Along every feasible curve through x^* , the derivative of f must vanish. Thus $\nabla f(x^*)$ is orthogonal to the tangent space, just as $\nabla g(x^*)$ is. Its normal space is one-dimensional, so the two gradients are collinear.

Define the Lagrangian $\mathcal{L}(x, \lambda) = f(x) + \lambda g(x)$. Solve stationarity in x together with the original constraint, then decide which candidates actually minimize. Multipliers give necessary conditions under the stated regularity assumption; they do not prove existence or global optimality by themselves.

Example: nearest point on a plane. Minimize $f = x^2 + y^2 + z^2$ under $g = x + y + z - 1 = 0$. The feasible plane is nonempty and closed, and f is coercive. Hence a minimizer exists. Since $\nabla g = (1, 1, 1) \neq 0$, its coordinates satisfy

$$2x + \lambda = 2y + \lambda = 2z + \lambda = 0, \quad x + y + z = 1.$$

Thus $x = y = z = 1/3$, $\lambda = -2/3$ and $f = 1/3$. For a direct global check, Cauchy-Schwarz gives $1 = (x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$, with equality only when all coordinates are equal.

Example: a rectangle. Minimize $f(a, b) = 2(a + b)$ for $a, b > 0$ under $g(a, b) = ab - A_0 = 0$. Stationarity gives $2 + \lambda b = 0$, $2 + \lambda a = 0$, so $a = b = \sqrt{A_0}$. The one-variable proof from Lab 3 supplies attainment and global minimality; the multiplier calculation recovers the same candidate.

4.5 Several equality constraints

Let $g = (g_1, \dots, g_m) : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, with $f, g_i \in C^1$. At a feasible point x^* , assume the gradients $\nabla g_1(x^*), \dots, \nabla g_m(x^*)$ are linearly independent (equivalently, $Dg(x^*)$ has row rank m). Then

$$T_{x^*}\{g = 0\} = \ker Dg(x^*).$$

If x^* is a local minimizer, $\nabla f(x^*)$ is orthogonal to this kernel and belongs to the span of the constraint gradients. Therefore

$$\nabla f(x^*) + \sum_{j=1}^m \lambda_j \nabla g_j(x^*) = 0, \quad g_1(x^*) = \dots = g_m(x^*) = 0.$$

There are $n + m$ scalar equations for n coordinates and m multipliers.

Worked example. Minimize $x^2 + y^2 + z^2$ under $x + y + z = 1$ and $x - y = 0$. The constraint gradients $(1, 1, 1)$ and $(1, -1, 0)$ are independent. The equations are

$$2x + \lambda + \mu = 0, \quad 2y + \lambda - \mu = 0, \quad 2z + \lambda = 0, \quad x + y + z = 1, \quad x = y.$$

They yield $x = y = z = 1/3$, $\mu = 0$ and $\lambda = -2/3$. The extra constraint is independent even though its multiplier is zero. Existence follows from closedness and coercivity; strict convexity on the affine feasible line proves uniqueness.

4.6 Check the assumptions, then check the candidate

A singular constraint. Minimize $f(x) = x$ under $g(x) = x^2 = 0$. The only feasible point is 0, so it is a minimizer. But $1 + \lambda 0 = 0$ has no solution: the constraint gradient vanishes.

A regular constraint with two kinds of candidates. For $f(x, y) = x$ on $x^2 + y^2 = 1$, multipliers give $(1, 0)$ and $(-1, 0)$. The former is the maximum and the latter the minimum.

Exit task. Before using multipliers, state the objective, the feasible set and the rank condition. After solving, explain why the chosen candidate is a minimizer.

5 Isoperimetry: two formulations and the polygonal route

By the end of the session: compare regular polygons at fixed area or perimeter, prove the equivalence of the two optimization problems, and pass from the polygonal inequality to the planar inequality.

5.1 Admissible shapes and the two questions

For these two sessions, a shape is the compact region enclosed by a rectifiable simple closed planar curve, with positive area. Write $A(K)$ for area and $P(K)$ for boundary length. Convex regions are an important subclass. Translations and dilations are allowed. We ask

$$\inf\{P(K) : A(K) = A_0\}, \quad \sup\{A(K) : P(K) = L\}, \quad A_0, L > 0.$$

We initially use infimum and supremum: attainment is part of the problem. The circle is a boundary curve; the corresponding region is a disk.

Session 2 stated the expected answer. We now build a proof, using the polygonal inequality from Lab 4. In particular, we will not use the planar inequality to prove the polygonal result on which our argument depends.

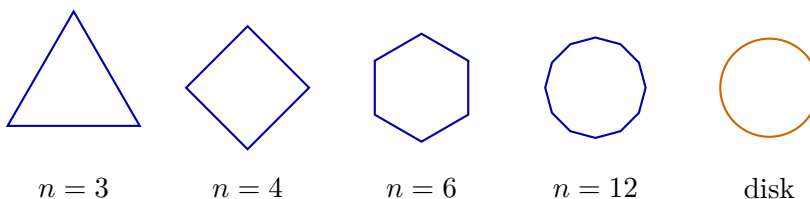
5.2 Regular polygons suggest the disk

A regular n -gon with side length s and apothem r has

$$P = ns, \quad r = \frac{s}{2 \tan(\pi/n)}, \quad A = \frac{1}{2}Pr.$$

Eliminating s, r gives the two useful formulas

$$\boxed{P_n(A_0) = 2\sqrt{A_0 n \tan(\pi/n)}}, \quad \boxed{A_n(L) = \frac{L^2}{4n \tan(\pi/n)}}.$$



All five regions have the same area; their perimeters decrease from left to right.

n	Perimeter when $A_0 = 1$	Area when $L = 1$
3	4.55901	0.04811
4	4.00000	0.06250
6	3.72242	0.07217
8	3.64072	0.07544
12	3.58630	0.07775
24	3.55509	0.07912
disk	3.54491	0.07958

Monotonicity, not just numerical evidence. Put $t = \pi/n$. Then $n \tan(\pi/n) = \pi \tan t/t$. For $0 < t < \pi/2$,

$$\left(\frac{\tan t}{t}\right)' = \frac{t - \sin t \cos t}{t^2 \cos^2 t} > 0,$$

because $h(t) = t - \sin t \cos t$ has $h(0) = 0$ and $h'(t) = 2 \sin^2 t > 0$. As n increases, t decreases, so $n \tan(\pi/n)$ strictly decreases to π . Therefore

$$P_n(A_0) \downarrow 2\sqrt{\pi A_0}, \quad A_n(L) \uparrow \frac{L^2}{4\pi}.$$

Every finite regular polygon has larger perimeter at fixed area, and smaller area at fixed perimeter, than the disk.

Discussion. Why do these comparisons still leave open the possibility of an irregular or nonpolygonal shape beating the disk?

5.3 Equivalence by rescaling competitors

For $t > 0$,

$$A(tK) = t^2 A(K), \quad P(tK) = tP(K).$$

Thus $P(K)^2/A(K)$, or its reciprocal, is invariant under scaling. Here is a proof of equivalence that does not assume the answer is a disk.

Suppose K_* maximizes area at perimeter L , with area $A_* > 0$. If a competitor H had area A_* and perimeter $p < L$, then $(L/p)H$ would have perimeter L and area $(L/p)^2 A_* > A_*$, a contradiction. Hence K_* also minimizes perimeter at area A_* .

Conversely, suppose K_* minimizes perimeter at area A_0 , with perimeter P_* . If H had perimeter P_* and area $a > A_0$, then $\sqrt{A_0/a} H$ would have area A_0 and perimeter smaller than P_* . This is again a contradiction. Dilating an optimizer transfers these statements to every positive target area or perimeter.

Equivalence does not itself give existence. It says that, in this scale-invariant admissible class, a solution of either problem yields solutions of the other. Fixed obstacles or a fixed containing box could invalidate this scaling argument.

5.4 Convexification: enlarge area and save boundary

Let $H = \text{conv } K$. We use the elementary convex-hull facts

$$A(H) \geq A(K), \quad P(H) \leq P(K).$$

Straight chords replace inward detours; this suggests the perimeter inequality. A proof using projections is given in the instructor notes. For a nonconvex Jordan region the area inequality is strict: its hull fills an open portion outside the original region.

At fixed perimeter L , taking the hull may leave unused boundary length. Set $t = L/P(H) \geq 1$ to restore the constraint. Then

$$P(tH) = L, \quad A(tH) = t^2 A(H) \geq A(K).$$

Consequently, it suffices to prove the desired bound for convex regions. In particular, any maximizer over our full class must be convex.

5.5 The polygonal approximation proof

Input from Lab 4. Every convex polygon Q with at most m vertices, $m \geq 3$, satisfies

$$4m \tan(\pi/m) A(Q) \leq P(Q)^2.$$

The existence of a polygonal optimizer and the exclusion of degenerate maximizers are part of the instructor proof for that lab. Merely solving the polygonal multiplier equations would not supply this input.

Approximation fact. Let K be a convex region of perimeter L . Choose m equally spaced boundary points in arc length, and let Q_m be their convex hull. Then

$$Q_m \subset K, \quad P(Q_m) \leq L, \quad A(Q_m) \rightarrow A(K).$$

For sufficiently large m the hull has positive area. Collinear sample points cause no problem: the hull has at most m vertices. Each chord is no longer than its corresponding arc, which proves the perimeter estimate. The missing boundary caps have areas tending to zero; the instructor notes give an explicit estimate.

Apply the polygonal inequality to this particular approximation of the arbitrary region K :

$$A(Q_m) \leq \frac{P(Q_m)^2}{4m \tan(\pi/m)} \leq \frac{L^2}{4m \tan(\pi/m)}.$$

Letting $m \rightarrow \infty$ proves $A(K) \leq L^2/(4\pi)$. For a general Jordan region, apply the convex result to its hull:

$$A(K) \leq A(\text{conv } K) \leq \frac{P(\text{conv } K)^2}{4\pi} \leq \frac{P(K)^2}{4\pi}.$$

We have now proved the planar isoperimetric inequality

$$\boxed{P(K)^2 \geq 4\pi A(K)}.$$

For this proof, convergence of the approximating perimeters is unnecessary; the upper bound $P(Q_m) \leq L$ is enough.

Attainment. A disk of radius $L/(2\pi)$ has perimeter L and area $L^2/(4\pi)$, so the supremum is attained. Equivalently, the minimum perimeter at area A_0 is $2\sqrt{\pi A_0}$, attained by a disk of radius $\sqrt{A_0/\pi}$.

Equality needs an additional argument. Strict polygonal inequalities can become equalities in the limit. The approximation proof above establishes the bound and disk attainment; it does not by itself exclude a different region attaining the same value. Session 6 supplies that geometric rigidity argument.

5.6 Dido's problem along a straight shore

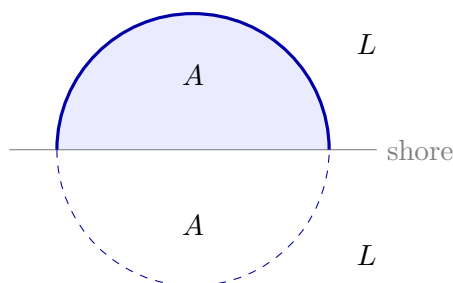
Model the shore by a straight line. A simple free arc of prescribed length L lies in one half-plane, meets the shore only at its endpoints, and encloses a region of area A together with the shore segment. The endpoints may move along the shore; the shore segment is not charged to L . Reflect the region across the shore. The doubled region has

$$\text{area } 2A, \quad \text{perimeter } 2L.$$

The planar inequality therefore gives

$$4\pi(2A) \leq (2L)^2, \quad \boxed{A \leq \frac{L^2}{2\pi}}.$$

A half-disk of radius L/π attains this bound. After the equality analysis in Session 6, it is also the only optimal shape, up to translation along the shore and choice of half-plane.



If the endpoints are prescribed, their separation is an additional constraint: an arbitrary fixed separation need not fit the half-disk. The free-endpoint problem is the one solved here.

Exit task. State the two ingredients in the approximation proof. Explain why comparing only regular polygons would not have been enough.

Reading. Blåsjö, *The Isoperimetric Problem*, pp. 528–529 and 542–543; Treibergs, *Inequalities that Imply the Isoperimetric Inequality*, Section 2. See the source map for the local PDFs and the Dido correction.

6 Steiner's geometric arguments and the existence gap

By the end of the session: use bisecting lines and hinge variations to characterize an optimizer, explain the cyclic-quadrilateral method, and identify exactly what an improvement argument does and does not prove.

6.1 A conditional argument: suppose an optimizer exists

Temporarily suppose K_* maximizes area among all our Jordan regions of perimeter L . We first reason as if existence were still unknown, to see the logical status of the classical geometric arguments. Session 5 will then supply the missing global result.

Convexification and rescaling imply that K_* is convex. It has no straight boundary segment: replace such a segment of length ℓ by two segments meeting a small distance h outward from its midpoint. The area increases by $\ell h/2$, while the added perimeter is

$$2\sqrt{(\ell/2)^2 + h^2} - \ell = O(h^2).$$

Rescaling to perimeter L costs only $O(h^2)$ in area, leaving a strict gain for small $h > 0$. This contradicts maximality. Hence K_* is strictly convex, meaning here that its boundary contains no line segment. This does not assert that its boundary is smooth.

6.2 Bisecting area and bisecting perimeter

Area bisectors exist. Fix a direction and translate a line perpendicular to it across K_* . The area on one side varies continuously from 0 to $A(K_*)$, so some position gives half the area. The line cuts the boundary at two points A, B and splits it into arcs of lengths ℓ_1, ℓ_2 , with $\ell_1 + \ell_2 = L$. If, for example, $\ell_1 < L/2$, reflect the first half-region across AB . The doubled region has the original area and perimeter $2\ell_1 < L$. Dilating to perimeter L increases area, a contradiction. Thus every area-bisecting line of an optimizer also bisects its perimeter.

Conversely, choose A on the boundary and let B be halfway around in arc length. The line AB cuts off two arcs of length $L/2$. If the two half-regions had unequal areas, reflecting the larger one would give perimeter L and area greater than $A(K_*)$. Thus every perimeter-bisecting chord also bisects area.

The length of the cutting chord is internal to the doubled construction; it is not part of its outer perimeter.

What reflection actually proves. We may replace a half by its reflection to obtain a symmetric optimizer. Equality of the two areas does not prove that the original halves were mirror images. We need an additional argument to determine their shapes.

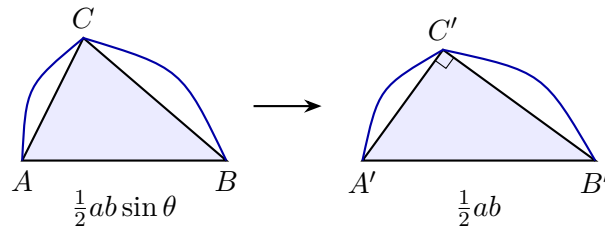
6.3 Steiner's hinge: a maximizing half must be a half-disk

Take one half-region bounded by a boundary arc from A to B and its chord AB . Its arc length is $L/2$ and its area is $A(K_*)/2$. Choose a point C on this arc, distinct from the endpoints. The half-region consists of triangle ACB and two caps between its sides AC, CB and the corresponding boundary arcs.

Treat the caps as rigid pieces attached to their chords, hinged at C . The two chord lengths $a = |AC|$ and $b = |BC|$ stay fixed. Change the angle $\theta = \angle ACB$, allowing A, B to slide on the cutting line. The boundary arcs keep their lengths and the caps keep their areas. The only changing area is

$$\frac{1}{2}ab \sin \theta.$$

If $\theta \neq \pi/2$, a sufficiently small motion toward $\pi/2$ strictly increases this area. Reflecting the improved half across its base gives a region with perimeter L and larger area, contradicting optimality.



The chords keep their lengths and the caps move rigidly. A small improving motion is enough; the right-angle position illustrates its target.

Admissibility matters. Use a small hinge motion so the new arc stays simple and above its base, with no overlapping caps. Strict convexity of the original region permits such a local motion; the instructor notes explain the separation near and away from the hinges. The competitor need not remain convex, since our admissible class includes all Jordan regions. One should not silently assume that an arbitrarily large flex is admissible.

It follows that $\angle ACB = 90^\circ$ for every point C of the arc. By the converse of Thales' theorem, the whole arc lies on the circle with diameter AB , and it is its semicircle on the chosen side. Apply the same reasoning to the other half. Both halves have the same diameter AB , so together they form a disk.

6.4 A second method: quadrilaterals of fixed side lengths

For a convex quadrilateral with consecutive side lengths a, b, c, d , semiperimeter $s = (a+b+c+d)/2$, and opposite angles α, γ , the general area formula is

$$Q^2 = (s-a)(s-b)(s-c)(s-d) - abcd \cos^2\left(\frac{\alpha+\gamma}{2}\right).$$

This is commonly called *Bretschneider's formula*. For a cyclic quadrilateral, $\alpha + \gamma = \pi$ and it reduces to *Brahmagupta's formula*

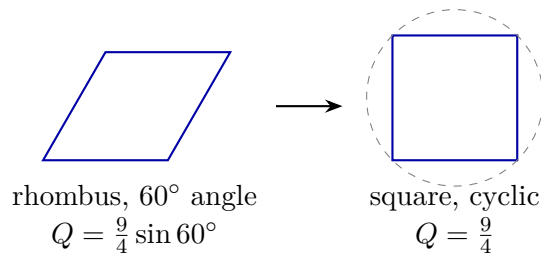
$$Q = \sqrt{(s-a)(s-b)(s-c)(s-d)}.$$

Treibergs presents the general formula in his Brahmagupta section. The name matters less than retaining the angle correction for a noncyclic quadrilateral.

For fixed feasible positive side lengths, the general formula gives

$$Q \leq \sqrt{(s-a)(s-b)(s-c)(s-d)},$$

with equality exactly when the quadrilateral is cyclic. A cyclic convex quadrilateral with those sides exists when each side is shorter than the sum of the other three. Lab 6 verifies both the maximum property and a local way of increasing area without assuming the planar isoperimetric inequality.



6.5 Four hinges on the boundary

Suppose the strictly convex optimizer K_* were not a disk. Choose three distinct boundary points; strict convexity makes them noncollinear, so they determine a unique circle. Some fourth boundary point must be off that circle, since otherwise the entire Jordan boundary would be that circle. Order the four points around the boundary to obtain a strictly convex noncyclic quadrilateral.

The region consists of this quadrilateral plus four caps attached to its sides. Flex the quadrilateral while keeping all four sides fixed and moving the caps rigidly. A sufficiently small area-increasing flex exists for a noncyclic quadrilateral. The four boundary arcs keep their lengths, the cap areas remain unchanged, and the central area strictly increases. The same local admissibility check used for the two-cap construction makes the new boundary simple. This contradicts maximality.

For precision, if t is one diagonal and $\alpha, \gamma \in (0, \pi)$ are the opposite angles facing it, then

$$\frac{dQ}{dt} = \frac{t}{2}(\cot \alpha + \cot \gamma) = \frac{t \sin(\alpha + \gamma)}{2 \sin \alpha \sin \gamma}.$$

This derivative is nonzero unless $\alpha + \gamma = \pi$. Therefore a small change of t in one direction increases area. This local argument avoids assuming that a full motion to a cyclic configuration preserves the attached boundary pieces without intersections.

6.6 What was missing, and how the polygonal proof supplies it

Both geometric methods initially prove a conditional statement:

if an area maximizer exists at perimeter L , it is a disk.

They do not by themselves prove that the supremum is attained. A sequence of improving shapes need not converge in a suitable sense, and an increasing sequence of area values need not reach its supremum. Even a stationary candidate plus a rule that improves every other candidate does not settle global optimality without an additional argument; Lab 6 gives an example.

Our logical order is now complete:

1. Lab 4 proves the polygonal inequality, including the existence argument for the finite-dimensional polygon problem.
2. Session 5 approximates an arbitrary convex region by its own inscribed polygons, proving $P^2 \geq 4\pi A$ for all admissible regions.
3. A disk attains the bound, so an optimizer really exists.
4. Either hinge argument shows that every optimizer is a disk.

Thus equality holds only for disks, up to rigid motions; the optimal size is fixed by the area or perimeter constraint. Reflection gives the half-disk equality case in the free-endpoint Dido problem.

Exit task. In a proposed isoperimetric proof, mark the first use of “let an optimizer be given.” Identify what establishes its existence.

Reading. Blåsjö, pp. 533–535 (Steiner’s hinge arguments and the discussion of existence); Treibergs, Sections 4–5, pp. 7–10 (quadrilateral formula and four hinges). The local-flex details in these notes make the geometric admissibility assumptions explicit.

7 Sequences of shapes: distances and compactness

By the end of the session: compare L^1 and Hausdorff convergence, use distance functions to describe compact sets, and prove a selection theorem using Arzelà–Ascoli.

7.1 What should it mean for shapes to converge?

A minimizing sequence needs a limit that remains admissible and controls the objective. We must specify what convergence means. For L^1 distance, use measurable subsets of a fixed bounded measurable region. For Hausdorff distance, use *nonempty compact* subsets of a fixed compact container $D \subset \mathbb{R}^d$. Limits may have empty interior.

7.2 L^1 distance: compare the occupied regions

The characteristic function of a set E is χ_E , equal to 1 on E and 0 outside it. Define

$$d_1(E, F) = \|\chi_E - \chi_F\|_{L^1} = \int |\chi_E - \chi_F| dx = |E \triangle F|.$$

Thus d_1 measures volume occupied by exactly one shape. Symmetry and the triangle inequality follow from the L^1 norm. But $d_1(E, F) = 0$ means equality *almost everywhere*. It is a metric only after identifying sets differing by a null set.

A disk and that disk with an isolated point have distance zero, as do a planar segment and the empty set. Slits of zero area are also invisible.

Area is continuous in this metric:

$$||E| - |F|| \leq |E \triangle F|.$$

This is useful, but neither boundary position nor perimeter is controlled by a small symmetric difference.

7.3 Hausdorff distance: compare point locations

For nonempty compact K , let

$$d_K(x) = \text{dist}(x, K) = \min_{y \in K} |x - y|.$$

The minimum exists by compactness. The directed excess $\rho(K, L) = \max_{x \in K} d_L(x)$ need not be symmetric. Define

$$d_H(K, L) = \max\{\rho(K, L), \rho(L, K)\}.$$

For $\varepsilon \geq 0$, set $K_\varepsilon = \{x : d_K(x) \leq \varepsilon\}$. Then

$$d_H(K, L) \leq \varepsilon \iff K \subset L_\varepsilon \text{ and } L \subset K_\varepsilon.$$

Every point in either set must have a nearby point in the other set. Controlling only one inclusion is not enough.

Metric properties. Nonnegativity and symmetry are immediate. If $d_H(K, L) = 0$, closedness gives both inclusions $K \subset L$ and $L \subset K$, hence $K = L$. If $d_H(K, L) \leq a$ and $d_H(L, M) \leq b$, choose successively nearby points in L and M . The ordinary triangle inequality gives $K \subset M_{a+b}$ and $M \subset K_{a+b}$, proving

$$d_H(K, M) \leq d_H(K, L) + d_H(L, M).$$

Reference convention. Henrot–Pierre also define convergence of open domains using $d_H(D \setminus \Omega_n, D \setminus \Omega)$, the distance between their compact complements. This differs from our ordinary compact-set metric: check the convention before transferring properties such as connectedness or diameter continuity. See Chapter 2, Sections 2.2.2–2.2.3 and 2.2.6.1, especially Theorem 2.2.25 and Proposition 2.2.27; also manuscript SH0_2025_Curs7.pdf, pp. 2–11.

7.4 The two distances detect different defects

A thin extension: L^1 convergence without Hausdorff convergence. Let $S = [0, 1]^2$ and

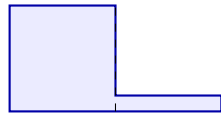
$$K_n = S \cup ([1, 2] \times [0, 1/n]).$$

These are connected compact regions, but $d_1(K_n, S) = 1/n \rightarrow 0$ while $d_H(K_n, S) = 1$. The extension has vanishing area but reaches a fixed distance from S .

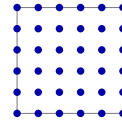
A fine grid: Hausdorff convergence without L^1 convergence. Let

$$G_n = \{(i/n, j/n) : i, j = 0, \dots, n\}.$$

Then $d_H(G_n, S) = \sqrt{2}/(2n) \rightarrow 0$, but $|G_n| = 0$ and $d_1(G_n, S) = 1$. Consequently, area is not Hausdorff-continuous on all compact sets.



small area, distant tip



dense points, zero area

Perimeter is not generally continuous either: the fine-toothed boundary from Session 2 converges in position to a straight edge, without its lengths converging to that edge's length.

7.5 Useful consequences of Hausdorff convergence

Suppose $K_n \rightarrow K$ in d_H .

- If $x_n \in K_n$ and $x_n \rightarrow x$, then $x \in K$. Indeed, $d_K(x) \leq |x - x_n| + d_H(K_n, K) \rightarrow 0$.
- Every $x \in K$ is a limit of points $x_n \in K_n$, chosen with $|x - x_n| \leq d_H(K_n, K)$.
- Inclusions pass to the limit: $K_n \subset L_n$, $L_n \rightarrow L$ imply $K \subset L$.
- Convexity passes to the limit: approximate $x, y \in K$ by $x_n, y_n \in K_n$, then pass to the limit in $(1 - t)x_n + ty_n \in K_n$ for each $t \in [0, 1]$.
- Diameter is continuous, even without convexity:

$$|\text{diam } K - \text{diam } L| \leq 2d_H(K, L).$$

Connectedness is also preserved for nonempty compact sets; a separation of the limit into two compact pieces a positive distance apart would separate all sufficiently close approximants. Intersections, however, need not commute with the limit; Lab 7 gives an example even for convex sets with nonempty intersections.

7.6 Distance functions turn sets into functions

Every d_K is 1-Lipschitz: $|d_K(x) - d_K(y)| \leq |x - y|$. For nonempty compact $K, L \subset D$,

$$d_H(K, L) = \sup_{x \in D} |d_K(x) - d_L(x)|.$$

To prove the upper bound, a nearest point in one set is at distance at most $d_H(K, L)$ from the other. Thus $d_K(x) \leq d_L(x) + d_H(K, L)$ and conversely. For the lower bound, evaluate the difference at points of K and of L , where one distance is zero. The same identity holds with the supremum over all $\mathbb{R}^d$; the bounded container suffices for the compactness argument.

7.7 Blaschke selection from Arzelà–Ascoli

Theorem 7.1 (Arzelà–Ascoli, the form needed here). *On a compact metric space D , a uniformly bounded equicontinuous sequence of continuous real-valued functions has a uniformly convergent subsequence. Equicontinuity means a common modulus of continuity for all functions in the sequence.*

The 1-Lipschitz bound supplies equicontinuity. The distance-function identity requires uniform, not merely pointwise, convergence.

Theorem 7.2 (Selection theorem for compact sets). *Let $D \subset \mathbb{R}^d$ be a fixed compact set. Every sequence of nonempty compact sets $K_n \subset D$ has a subsequence converging in Hausdorff distance to a nonempty compact set $K \subset D$. If all K_n are convex, their limit is convex. This last assertion is the usual convex form of Blaschke’s selection theorem; the compact-set version is stronger.*

Proof. Put $f_n = d_{K_n}|_D$. Then $0 \leq f_n \leq \text{diam } D$, and the f_n are all 1-Lipschitz. Arzelà–Ascoli gives, after passing to a subsequence, $f_n \rightarrow f$ uniformly on D . The limit is nonnegative and 1-Lipschitz. Define $K = \{x \in D : f(x) = 0\}$.

First K is compact and nonempty. It is closed in D ; choose $x_n \in K_n$ and a further subsequence with $x_n \rightarrow x \in D$. Uniform convergence and the Lipschitz bound give $f(x) = 0$.

We must show that f is actually a distance function. For $y \in K$, the Lipschitz property gives $f(x) \leq |x - y|$, hence $f(x) \leq d_K(x)$. For each fixed $x \in D$, choose nearest points $y_n \in K_n$. Along a further subsequence $y_n \rightarrow y \in D$. Uniform convergence gives $f(y) = 0$ and

$$f(x) = \lim_n d_{K_n}(x) = \lim_n |x - y_n| = |x - y| \geq d_K(x).$$

Thus $f = d_K$ on D . The distance-function identity now yields $d_H(K_n, K) \rightarrow 0$ along the original uniformly convergent subsequence. Convexity of the limit follows from the earlier segment argument. $\square$

The further subsequence of nearest points may depend on the fixed x ; it proves the identity at that point. It does not replace the common uniformly convergent subsequence of functions.

What compactness has not yet proved. Translating a unit disk to $(n, 0)$ prevents a convergent subsequence: the common compact container is essential. Also, $[0, 1] \times [0, 1/n]$ converges to a segment. Nonempty interior needs an additional nondegeneracy constraint to survive the limit.

Blaschke gives a compact limit, not automatically an admissible optimizer. Session 8 checks closedness of constraints and continuity of functionals.

Exit task. In the selection proof, explain why defining K as the zero set of f is not the final step. What still needs to be shown?

8 Existence for convex shapes: support functions and width

By the end of the session: apply the direct method in the Hausdorff metric, represent convex sets by support functions, prove continuity of geometric functionals, and solve area minimization at prescribed minimum width.

8.1 An existence theorem on a space of shapes

Let $\mathcal{K}(D)$ be the nonempty compact subsets of a fixed compact container D . By Session 7, this is a compact metric space for d_H .

Theorem 8.1 (Direct method for Hausdorff convergence). *If $\mathcal{A} \subset \mathcal{K}(D)$ is nonempty and closed for Hausdorff convergence, and $J : \mathcal{A} \rightarrow \mathbb{R}$ is continuous, then J attains both its minimum and maximum. For minimization, continuity can be weakened to lower semicontinuity, provided the infimum is finite.*

Proof. For a finite infimum m , choose $K_n \in \mathcal{A}$ with $J(K_n) \rightarrow m$. Blaschke gives a subsequence $K_{n_j} \rightarrow K$. Closedness puts K in $\mathcal{A}$. Lower semicontinuity gives $m \leq J(K) \leq \liminf_j J(K_{n_j}) = m$. A continuous real-valued function on the compact class is bounded, so its infimum is finite; apply the same argument to $-J$ for a maximum. $\square$

The practical questions are: what supplies a common container, which constraints survive the limit, and how does the objective behave? Translation-invariant problems often require translating the sequence before applying the theorem.

8.2 Support functions describe compact convex sets

For a nonempty compact convex $K \subset \mathbb{R}^d$ and a unit vector u , define

$$h_K(u) = \max_{x \in K} x \cdot u.$$

The supporting hyperplane with outward normal u is $\{x : x \cdot u = h_K(u)\}$. This is a *signed* position relative to the chosen origin; h_K can be negative if the origin is outside K . It is not always an unsigned distance.

In the plane write $u_\theta = (\cos \theta, \sin \theta)$ and $h_K(\theta) = h_K(u_\theta)$. Examples are

$$h_{\overline{B}(c,r)}(u) = c \cdot u + r, \quad h_{[-a,a] \times [-b,b]}(\theta) = a|\cos \theta| + b|\sin \theta|.$$

For a polygon with vertices v_i , $h_K(u) = \max_i v_i \cdot u$. Translation and dilation give

$$h_{K+a}(u) = h_K(u) + a \cdot u, \quad h_{tK}(u) = th_K(u) \quad (t \geq 0).$$

If $K \subset \overline{B}(0, R)$, then $|h_K(u) - h_K(v)| \leq R|u - v|$.

The support function determines K :

$$K = \bigcap_{|u|=1} \{x : x \cdot u \leq h_K(u)\}.$$

To see the reverse inclusion, take $x \notin K$ and a nearest point $y \in K$. Convexity implies $(x - y) \cdot (z - y) \leq 0$ for all $z \in K$ (differentiate squared distance along the segment from y to z). Thus $u = (x - y)/|x - y|$ separates x from K : $x \cdot u > y \cdot u = h_K(u)$. For a nonconvex compact set the support function determines its convex hull, so information about holes and inward dents is lost.

8.3 Hausdorff distance equals uniform support-function distance

Theorem 8.2. *For nonempty compact convex sets K, L ,*

$$d_H(K, L) = \max_{|u|=1} |h_K(u) - h_L(u)|.$$

In particular, Hausdorff convergence is equivalent to uniform convergence of support functions on the unit sphere.

Proof. If $d_H(K, L) = \varepsilon$, every $x \in K$ has a point $y \in L$ with $|x - y| \leq \varepsilon$. Hence $h_K(u) \leq h_L(u) + \varepsilon$, and the symmetric inequality gives the upper bound for the support difference.

Conversely, let the uniform support difference be δ . For $x \in K \setminus L$, take a nearest point $y \in L$ and the separating normal u above. Then

$$\text{dist}(x, L) = x \cdot u - h_L(u) \leq h_K(u) - h_L(u) \leq \delta.$$

The same argument with the sets exchanged gives $d_H(K, L) \leq \delta$. □

8.4 Width, diameter and constant width

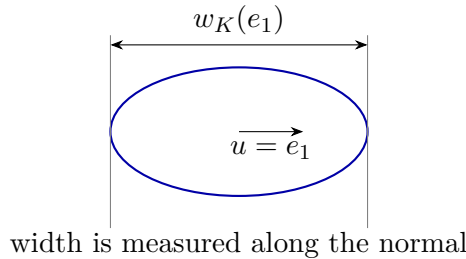
The width in direction u is the distance between the two supporting hyperplanes perpendicular to u :

$$w_K(u) = h_K(u) + h_K(-u).$$

In the plane, $w_K(\theta) = h_K(\theta) + h_K(\theta + \pi)$. The minimum width and maximum width are

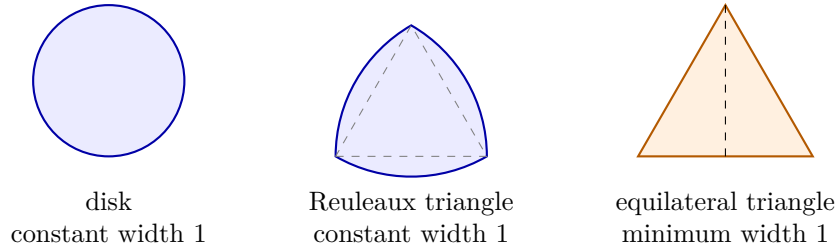
$$w(K) = \min_{|u|=1} w_K(u), \quad \max_{|u|=1} w_K(u) = \text{diam } K.$$

For the second identity, any projection difference is at most a distance between two points, while a pair realizing the diameter realizes that projection difference in its own direction.



A body has *constant width* w_0 when $w_K(u) = w_0$ in every direction. A disk of radius $w_0/2$ has this property, but it is not the only example: intersect three closed disks of radius w_0 whose centers form an equilateral triangle of side w_0 to obtain a Reuleaux triangle, also of constant width w_0 .

Minimum width w_0 is a different constraint: all directional widths are at least w_0 , with equality in at least one direction. An equilateral triangle of altitude w_0 has minimum width w_0 and diameter $2w_0/\sqrt{3}$; it is not a constant-width body.



8.5 Which functionals become continuous?

If $d_H(K, L) \leq \varepsilon$ for compact convex sets, then $\|w_K - w_L\|_\infty \leq 2\varepsilon$. Taking minima or maxima gives

$$|w(K) - w(L)| \leq 2\varepsilon, \quad |\text{diam } K - \text{diam } L| \leq 2\varepsilon.$$

Thus a prescribed minimum width or prescribed constant width is a closed constraint.

Area. If planar compact convex $K_n \rightarrow K$, then $|K_n \triangle K| \rightarrow 0$, hence $A(K_n) \rightarrow A(K)$. Here is the idea when K contains a disk $\overline{B}(z, r)$, $r > 0$. Translate z to zero. If $d_H(K_n, K) = \varepsilon < r$, then $h_K \geq r$ and support-function comparison gives

$$(1 - \varepsilon/r)K \subset K_n \subset (1 + \varepsilon/r)K.$$

The area between these two dilates tends to zero and contains the symmetric difference. If the convex limit is a segment or a point, then $K_n \subset K_\varepsilon$ and the area of this thin neighborhood tends to zero. This also handles degeneration. The volume version holds in every dimension.

Perimeter in the plane. Cauchy's projection formula is

$$P(K) = \int_0^\pi w_K(\theta) d\theta.$$

For a convex polygon, integrate its total projected boundary variation: each side of length ℓ contributes 2ℓ , and the projected closed boundary has variation twice its width. Approximation gives the formula for convex bodies. Consequently

$$|P(K) - P(L)| \leq 2\pi d_H(K, L).$$

For compactness arguments we extend this formula to degenerate convex sets: a segment of length ℓ has $P = 2\ell$ and a point has $P = 0$. This continuous extension is not the ordinary one-dimensional measure of the boundary of a segment, which is ℓ .

Functional on planar compact convex sets	Continuity mechanism
Area; integral $\int_K f$ for bounded continuous f	L^1 convergence of indicators
Perimeter	Cauchy's formula and uniform widths
Minimum width and diameter	Minimum/maximum of uniformly convergent widths
Inradius and circumradius	Continuous; each is 1-Lipschitz in d_H (instructor notes)

Area and perimeter continuity here rely on convexity. Diameter continuity already holds for arbitrary nonempty compact sets.

8.6 Two direct-method applications

Isoperimetric existence, independently of identification. Consider compact convex sets with extended perimeter $L > 0$. Their diameters are at most $L/2$: two boundary arcs join any diameter pair, each of length at least that diameter. Translate one point of each set to the origin, confining the sets to $\overline{B}(0, L/2)$. Perimeter is continuous, so the constraint survives the limit. Area is continuous, so a maximizing sequence has a maximizing limit. A disk competitor has positive area; the maximizer therefore is a body, not a segment. The geometric arguments of Sessions 5–6 identify it.

Area optimization at constant width. Constant width $w_0 > 0$ implies diameter w_0 , so translation gives a common container. Uniform support-function convergence preserves all the equalities $w_K(u) = w_0$. Such a limit has positive minimum width and hence interior. Area therefore attains a minimum and maximum in this class. This proves existence without identifying either optimizer. In particular, the equilateral triangle is not a candidate in this class.

8.7 Area minimization at prescribed minimum width

We study

$$\inf\{A(K) : K \subset \mathbb{R}^2 \text{ compact convex, } w(K) \geq w_0\}, \quad w_0 > 0.$$

The condition $w(K) \geq w_0$ does not bound diameter: rectangles $[0, n] \times [0, w_0]$, for $n \geq w_0$, are a counterexample. Instead use the *objective sublevel* to supply confinement.

Diameter bound from area and width. Take a diameter segment AB of length d and support points on either side of its line, at heights $h_+, h_- \geq 0$. The two triangles with base AB lie in K , have disjoint interiors, and satisfy $h_+ + h_- \geq w(K)$. Hence

$$A(K) \geq \frac{1}{2} \text{diam}(K)w(K).$$

An equilateral triangle of altitude w_0 is an admissible competitor with area $A_* = w_0^2/\sqrt{3}$. Choose a minimizing sequence with areas at most $2A_*$. Its diameters are at most $4w_0/\sqrt{3}$. Translate a point of each set to the origin and apply Blaschke. The limit is convex and still has width at least w_0 ; area continuity gives a minimizer. If its minimum width were greater than w_0 , scaling down to width w_0 would reduce area. Thus the minimum also solves the equality-constrained problem $w(K) = w_0$.

Theorem 8.3 (Pál’s minimum-width inequality). *For every planar convex body,*

$$A(K) \geq \frac{w(K)^2}{\sqrt{3}}.$$

Equality holds exactly for equilateral triangles. Therefore the unique optimal shape at prescribed minimum width w_0 is an equilateral triangle of altitude w_0 , up to rigid motions, with area $w_0^2/\sqrt{3}$ and side length $2w_0/\sqrt{3}$.

8.8 Incircle proof: geometry followed by one-variable calculus

Normalize to $w(K) = 1$ and choose a largest inscribed disk, of radius r . The incircle contact lemma from Yaglom–Boltyanskii gives

$$\frac{1}{3} \leq r \leq \frac{1}{2}.$$

If there are two antipodal contact points, their common supporting strip has width $2r = 1$, and $A(K) \geq \pi/4 > 1/\sqrt{3}$. Otherwise three contact tangents enclose a triangle containing K and having the same incircle. The instructor notes prove the contact lemma and the radius bounds.

For $1/3 \leq r < 1/2$, choose a farthest support point opposite each contact tangent. Each lies at distance at least $1 - r$ from the incircle center. Shorten these three radial segments to length $1 - r$. Convexity keeps their endpoints inside K , along with the disk and the tangent regions from each endpoint to the disk. These three regions outside the disk occupy separate sectors, so their areas may be added. If

$$\alpha = \arccos\left(\frac{r}{1-r}\right),$$

each added area is $r\sqrt{1-2r} - r^2\alpha$. Therefore

$$A(K) \geq F(r) := \pi r^2 + 3r\sqrt{1-2r} - 3r^2 \arccos\left(\frac{r}{1-r}\right).$$

Lab 8 checks the area calculation and the identity

$$F'(r) = 6r \left[\frac{\pi}{3} - \arccos\left(\frac{r}{1-r}\right) \right] + \frac{3(1-2r)^{3/2}}{1-r} > 0 \quad (1/3 < r < 1/2).$$

Thus $A(K) \geq F(1/3) = 1/\sqrt{3}$. Equality requires $r = 1/3$; the equality case of the incircle bound forces an equilateral triangle. Scaling restores arbitrary $w(K) > 0$.

What is supplied in class. The incircle contact lemma and the separation of the three tangent regions are stated geometric inputs for the guided lab. Their proofs are in the instructor notes. This lets the course focus on convergence and existence without hiding the work needed to identify the optimizer.

Exit task. Explain why width-constraint closedness alone does not prove existence, and why minimum width and constant width give different admissible classes.

Reading. Manuscript SH0_2025_Curs8.pdf, pp. 3–16; Yaglom–Boltyanskii, Chapter 6, Exercises 6-2 and 6-4 and their solutions (printed pp. 215–217 and 221–222); Henrot–Pierre, Chapter 2, for the direct-method framework and topology conventions.

9 Mixed areas, mixed volumes and Brunn–Minkowski

By the end of the session: compute with Minkowski sums, interpret mixed area as a first variation, prove Brunn–Minkowski by induction on unions of boxes, and deduce the isoperimetric inequality.

9.1 Adding shapes

For nonempty compact sets in $\mathbb{R}^d$, define

$$K + L = \{x + y : x \in K, y \in L\}, \quad tK = \{tx : x \in K\} \quad (t \geq 0).$$

The sum is compact; if both sets are convex, so is the sum. Addition is commutative and associative. For convex K , $sK + tK = (s + t)K$; the reverse inclusion uses convexity and need not hold without it. This is a sum of all pairs of points, not a union or an operation on corresponding vertices alone.

Support functions make the operation simple:

$$h_{K+L}(u) = h_K(u) + h_L(u), \quad h_{tK}(u) = th_K(u).$$

Indeed, the two maximizations of $(x + y) \cdot u$ are independent. For compact convex sets this also proves

$$d_H(K + L, K' + L') \leq d_H(K, K') + d_H(L, L').$$

Thus the continuity results from Session 8 apply to sums of converging convex shapes.

Two useful models are a safety margin $K + tB$, where B is the closed unit ball, and the forbidden positions $O + (-R)$ of a translating robot R around an obstacle O . The latter follows because $(z + R) \cap O \neq \emptyset$ exactly when $z = o - r$ for some $o \in O, r \in R$. The minus sign reflects the robot; it does not mean set subtraction.

The *difference body* satisfies $h_{K+(-K)}(u) = h_K(u) + h_K(-u) = w_K(u)$. Consequently a convex body has constant width w if and only if $K + (-K) = wB$. The ball here has radius w , not $w/2$.

9.2 Mixed area: the cross term in an area polynomial

For planar compact convex sets, area is quadratic under Minkowski addition:

$$A(sK + tL) = s^2 A(K) + 2st V(K, L) + t^2 A(L), \quad s, t \geq 0. \quad (1)$$

The coefficient $V(K, L)$ is called the *mixed area*. In particular,

$$V(K, L) = \frac{A(K + L) - A(K) - A(L)}{2}, \quad \left. \frac{d}{dt} A(K + tL) \right|_{t=0+} = 2V(K, L).$$

It measures the initial area growth when adding a small copy of L . It is not the area of $K \cap L$.

Why a quadratic? For polygons, merge the two cyclic lists of edge directions. In this common list, allow zero edges when a direction is absent. The vertices of $sK + tL$ are corresponding sums $sp_i + tq_i$. Substituting them in the shoelace formula gives a homogeneous polynomial of degree two. Its pure coefficients are $A(K)$ and $A(L)$. Approximate general convex sets by polygons and use Hausdorff continuity of sums and area. The instructor notes give the details.

Example: aligned rectangles. If $K = [0, a] \times [0, b]$ and $L = [0, c] \times [0, d]$, then

$$A(K + tL) = (a + tc)(b + td) = ab + t(ad + bc) + t^2 cd, \quad V(K, L) = \frac{1}{2}(ad + bc).$$

The mixed area has units of area, like both other coefficients.

Mixed area is symmetric, translation invariant, continuous and additive in each argument for Minkowski addition. It is homogeneous of degree one in each argument, and $V(K, K) = A(K)$. For a polygon K with edge lengths ℓ_i and outward unit normals u_i ,

$$2V(K, L) = \sum_i \ell_i h_L(u_i). \quad (2)$$

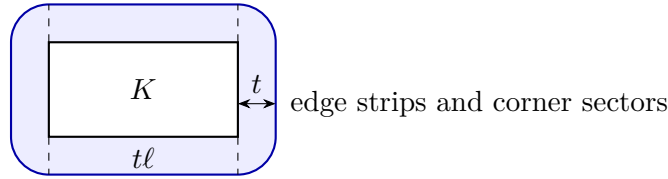
The supporting line at edge i moves outward by $th_L(u_i)$; edge strips give the linear area term, while corner contributions are of order t^2 . Translate L to contain the origin for this argument. The formula is independent of that translation because $\sum_i \ell_i u_i = 0$. Support-function comparison shows that $L \subset M$ implies $V(K, L) \leq V(K, M)$; symmetry gives monotonicity in the other argument.

9.3 Parallel bodies and the planar Steiner formula

For a planar convex body K , put $K_t = K + tB$, where B is the unit disk. Then

$$A(K_t) = A(K) + tP(K) + \pi t^2, \quad P(K_t) = P(K) + 2\pi t. \quad (3)$$

For a convex polygon, the added region consists of one strip of area $t\ell_i$ along each edge and circular sectors at the vertices. Their exterior angles sum to 2π , so the sectors total πt^2 . The parallel boundary consists of the original side lengths and arcs with total length $2\pi t$. Convex polygonal approximation, together with Session 8's continuity results, gives the formula for convex bodies.



Comparison with the mixed-area polynomial gives

$$V(K, B) = P(K)/2, \quad V(B, B) = \pi.$$

Convexity is essential to this exact formula. For two disjoint disks whose small parallel neighborhoods remain disjoint, the quadratic term is $2\pi t^2$, not πt^2 .

9.4 Brunn–Minkowski: a statement about volume growth

Write $|K|$ for d -dimensional volume in this subsection.

Theorem 9.1 (Brunn–Minkowski). *For nonempty compact sets $K, L \subset \mathbb{R}^d$,*

$$|K + L|^{1/d} \geq |K|^{1/d} + |L|^{1/d}. \quad (4)$$

Equivalently, for $0 \leq \lambda \leq 1$,

$$|(1 - \lambda)K + \lambda L|^{1/d} \geq (1 - \lambda)|K|^{1/d} + \lambda|L|^{1/d}.$$

For convex bodies with positive volume, equality at $0 < \lambda < 1$ holds exactly when $L = aK + z$ for some $a > 0$ and vector z .

The weighted form follows by applying (4) to the dilated sets. For convex sets, it says that the d th root of volume is concave along Minkowski interpolation. The equality characterization is a separate rigidity theorem, stated here without proof; the box induction below proves the inequality. Homothetic sets do give equality directly.

9.5 The inductive proof with unions of boxes

A box is a product of closed intervals of positive lengths, with edges parallel to the coordinate axes. Boundaries have volume zero. Represent each finite union by boxes with pairwise disjoint interiors, subdividing at their face coordinates if necessary. Induct on the total number $N = m + n$ of boxes in the two representations.

Base case: one box in each set. If their side lengths are $a_i, b_i > 0$, the sum has side lengths $a_i + b_i$. By the ordinary arithmetic–geometric mean inequality,

$$\left(\prod_{i=1}^d \frac{a_i}{a_i + b_i} \right)^{1/d} + \left(\prod_{i=1}^d \frac{b_i}{a_i + b_i} \right)^{1/d} \leq \frac{1}{d} \sum_{i=1}^d \frac{a_i + b_i}{a_i + b_i} = 1.$$

Multiply by $(\prod_i (a_i + b_i))^{1/d}$ to obtain (4). In the plane, this is just $(a + c)(b + d) \geq (\sqrt{ab} + \sqrt{cd})^2$.

Induction step: cut in matching volume proportions. Suppose $m \geq 2$, exchanging the sets if needed. Two boxes in K have disjoint interiors, so some coordinate hyperplane $x_i = c$ separates them. Cut all of K there into K^-, K^+ , both of positive volume, and put $\theta = |K^-|/|K| \in (0, 1)$. On each side retain only clipped boxes of positive volume. Omitting boundary-only pieces preserves the volume fractions and keeps the retained sets inside K .

Move a parallel hyperplane $x_i = e$ through L until it cuts off the same fraction: $|L^-| = \theta|L|$. Such a cut exists because the volume to one side varies continuously from zero to $|L|$. The positive-side fractions are then $1 - \theta$ for both sets.

Each of K^- and K^+ uses at most $m - 1$ nondegenerate boxes; each of L^-, L^+ uses at most n . A cut can split other boxes, but *each individual side* loses at least one of the two separated boxes of K . This is what makes both applications of induction valid.

The sums $K^- + L^-$ and $K^+ + L^+$ lie on opposite sides of $x_i = c + e$. Their overlap has volume zero, and both are contained in $K + L$. Thus, writing $a = |K|$ and $b = |L|$,

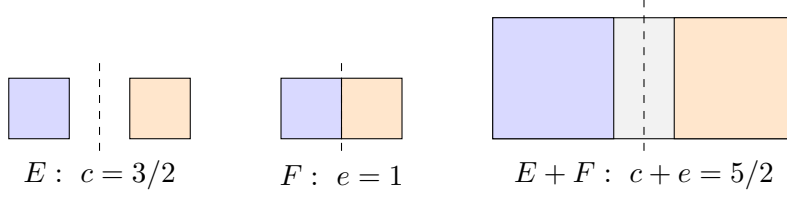
$$\begin{aligned} |K + L| &\geq |K^- + L^-| + |K^+ + L^+| \\ &\geq ((\theta a)^{1/d} + (\theta b)^{1/d})^d + (((1 - \theta)a)^{1/d} + ((1 - \theta)b)^{1/d})^d \\ &= \theta(a^{1/d} + b^{1/d})^d + (1 - \theta)(a^{1/d} + b^{1/d})^d \\ &= (a^{1/d} + b^{1/d})^d. \end{aligned}$$

No statement that these two sums exhaust $K + L$ is needed: the cross sums may supply additional points.

Pass from boxes to compact sets. Cover K by the finitely many closed dyadic grid cubes of side 2^{-j} that meet it, and call their union K_j ; similarly define L_j . If $\delta_j = \sqrt{d} 2^{-j}$,

$$K \subset K_j \subset K + \delta_j B, \quad K + L \subset K_j + L_j \subset K + L + 2\delta_j B.$$

For a compact set E , its closed ε -neighborhoods decrease to E , so finite-volume continuity from above gives $|E + \varepsilon B| \rightarrow |E|$. The inclusions therefore imply $|K_j| \rightarrow |K|$, $|L_j| \rightarrow |L|$ and $|K_j + L_j| \rightarrow |K + L|$. Pass to the limit in the box inequality. This uses *outer* approximation, not general volume continuity for Hausdorff convergence. The grid counterexample from Session 7 remains valid.



In this Lab 9 example, both cuts bisect area. The two colored sums have areas $4 + 4$, while the full sum has area 10. The extra band helps the inequality; the induction does not need to account for it.

9.6 Mixed-area inequality and planar isoperimetry

Apply Brunn–Minkowski to $K + tL$ and compare with (1):

$$A(K) + 2tV(K, L) + t^2A(L) \geq (\sqrt{A(K)} + t\sqrt{A(L)})^2.$$

Cancel the constant and quadratic terms and divide by $2t$, for $t > 0$:

$$\boxed{V(K, L) \geq \sqrt{A(K)A(L)}}. \quad (5)$$

For bodies of positive area, equality is equivalent to equality in Brunn–Minkowski, hence to positive homothety, using the stated rigidity theorem. In particular, setting $L = B$ gives

$$\frac{P(K)}{2} \geq \sqrt{\pi A(K)}, \quad \boxed{P(K)^2 \geq 4\pi A(K)}.$$

A disk attains the bound. The disk equality classification was also established in Sessions 5–6. This is a new proof of the inequality from volume growth, independent of the earlier polygonal argument. It proves the convex case directly; for rectifiable planar Jordan regions, the convex-hull reduction from Session 5 extends the bound.

A directional boundary cost. For a prescribed convex body L containing the origin in its interior, define $P_L(K) = 2V(K, L)$. For polygons, (2) weights each side by $h_L(u_i)$. Then $P_L(K) \geq 2\sqrt{A(K)A(L)}$. At fixed area A_0 , the minimizers are translated copies of $\sqrt{A_0/A(L)}L$, by the same rigidity statement. A noncircular cost therefore selects a noncircular shape.

9.7 Mixed volumes: the same idea in higher dimension

The volume-polynomial theorem for convex bodies generalizes the planar calculation. We state it; a full proof is beyond this 90-minute session. For $K, L \subset \mathbb{R}^d$ compact convex,

$$|K + tL| = \sum_{j=0}^d \binom{d}{j} V(K[d-j], L[j]) t^j.$$

The notation $K[d-j]$ means $d-j$ repetitions of K in a symmetric function of d arguments. The mixed volume is continuous, nonnegative, translation invariant, and Minkowski additive and positively homogeneous in each argument. Its diagonal value is $V(K[d]) = |K|$.

For boxes of side lengths (a, b, c) and (p, q, r) in $\mathbb{R}^3$,

$$|K + tL| = (a + tp)(b + tq)(c + tr), \quad V(K, K, L) = \frac{pbc + aqc + abr}{3}.$$

In particular, the linear coefficient is *three times* the first mixed volume. In dimension d it is $dV(K[d-1], L)$.

For $|K| > 0$, differentiate the Brunn–Minkowski lower bound at $t = 0+$:

$$V(K[d-1], L) \geq |K|^{(d-1)/d} |L|^{1/d}.$$

Taking $L = B$, put $\kappa_d = |B|$ and use the first variation of volume under an outer parallel displacement,

$$S(K) = \frac{d}{dt} |K + tB| \Big|_{0+} = dV(K[d-1], B).$$

Here S is surface area; this standard convex-body identity is supplied in higher dimension and was proved above in dimension two. We obtain

$$\boxed{S(K) \geq d \kappa_d^{1/d} |K|^{(d-1)/d}.$$

The ball attains equality. In $\mathbb{R}^3$, this is $S^3 \geq 36\pi|K|^2$. The higher-dimensional ball rigidity follows from the equality case of Minkowski's first inequality; the instructor notes explain its reduction to the stated Brunn–Minkowski rigidity theorem.

Exit task. Why do we cut in matching volume proportions? Why is the factor d necessary in the volume derivative?

Reading. Treibergs, Sections 7–8, pp. 14–18; Schneider, Second Expanded Edition (2014), Sections 4.2, 5.1, 7.1–7.2, especially Theorems 5.1.7, 7.1.1 and 7.2.1. For the manuscript introduction to sums and mixed areas, see SH0_2025_Curs9.pdf, pp. 8–10.

10 Constant width and the Blaschke–Lebesgue theorem

By the end of the session: recognize constant width through support functions and Minkowski addition, justify existence, and prove the sharp area bound using mixed areas.

10.1 A constraint in every direction

Throughout, B is the closed unit disk, of radius one, and $w > 0$. For a compact convex body K , write $u(\theta) = (\cos \theta, \sin \theta)$ and $h(\theta) = h_K(u(\theta))$. The following are equivalent:

$$w_K(u) = w \quad \text{for every unit } u; \quad h(\theta) + h(\theta + \pi) = w; \quad K + (-K) = wB.$$

Indeed, $h_{-K}(u) = h_K(-u)$, support functions add under Minkowski addition, and they determine compact convex sets. These equivalences also hold in $\mathbb{R}^d$. The disk of width w is $(w/2)B$. The difference body is unchanged by translation of K .

A body of constant width w has diameter w . For any $x, y \in K$, project onto $(x - y)/|x - y|$ to obtain $|x - y| \leq w$; the opposite supports in any direction have distance w . If their contact points are p, q , then $(p - q) \cdot u = w$ and $|p - q| \leq w$, forcing $p - q = wu$. This also proves that each support has just one contact point: two different choices of p cannot both equal $q + wu$. Thus constant-width bodies have no straight boundary segments, though they may have corners.

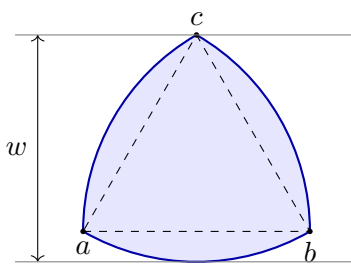
Reuleaux triangle. Let a, b, c form an equilateral triangle of side w , and set

$$R = \overline{B}(a, w) \cap \overline{B}(b, w) \cap \overline{B}(c, w).$$

Its boundary consists of three circular arcs, each of radius w and central angle $\pi/3$. In every pair of opposite support directions, one contact is a vertex and the other lies on the arc centered there. The support distance is therefore w . Counting sectors gives

$$P(R) = \pi w, \quad A(R) = 3 \frac{\pi w^2}{6} - 2 \frac{\sqrt{3} w^2}{4} = \frac{\pi - \sqrt{3}}{2} w^2.$$

The interior corner angles are $2\pi/3$.



10.2 Perimeter, symmetry and smooth examples

Cauchy's perimeter formula from Session 8 immediately gives

$$P(K) = \int_0^\pi w_K(u(\theta)) \, d\theta = \pi w. \quad (6)$$

This is *Barbier's theorem*. Alternatively, use the additivity of convex perimeter in $K + (-K) = wB$: $2P(K) = 2\pi w$. Consequently, minimizing perimeter in this class cannot distinguish shapes. If K is centrally symmetric, translate its center to zero. Then $K = -K$, so $2K = wB$: the only centrally symmetric example is the disk.

Write $h = w/2 + f$. Constant width becomes $f(\theta + \pi) = -f(\theta)$. For a trigonometric polynomial this means that only odd frequencies occur:

$$h(\theta) = \frac{w}{2} + \sum_{j \text{ odd}} (a_j \cos j\theta + b_j \sin j\theta).$$

The frequency-one terms translate the body; they do not change its shape. There is a second requirement: *the function must be a support function*. For a C^2 , periodic h , the planar criterion is

$$\rho = h + h'' \geq 0.$$

Indeed, the boundary parametrization and its derivative are

$$x = hu + h'u^\perp, \quad x' = \rho u^\perp, \quad u^\perp = (-\sin \theta, \cos \theta).$$

Nonnegative ρ makes the tangent turn in the convex orientation; the support-line verification is supplied in the instructor notes. For a general support function, the same criterion is distributional; we use that formulation only as further reading.

Example. $h(\theta) = w/2 + a \cos 3\theta$ gives $\rho = w/2 - 8a \cos 3\theta$. It defines a convex body precisely when $|a| \leq w/16$. The width identity holds even for larger a , but those functions are not support functions. Green's area formula gives

$$A(K) = \frac{1}{2} \int_0^{2\pi} h(h + h'') d\theta = \frac{1}{2} \int_0^{2\pi} (h^2 - (h')^2) d\theta. \quad (7)$$

For this example, $A = \pi w^2/4 - 4\pi a^2$. Smooth examples approach the edge of the convexity constraint when area decreases; a global minimizer need not have a smooth boundary.

10.3 Existence before identification

Proposition 10.1. *Among planar convex bodies of constant width w , area has a minimum and a maximum. More generally, any translation-invariant functional which is lower semicontinuous for Hausdorff convergence has a minimizer on this class, provided its infimum is finite.*

Proof. The disk makes the class nonempty. Translate one point of each member of a minimizing sequence to zero. Diameter w places the sequence in $\overline{B}(0, w)$. Blaschke selection gives a compact convex limit. Uniform support convergence preserves $h(u) + h(-u) = w$. The limit has interior: a lower-dimensional convex set has width zero in some direction. Lower semicontinuity finishes the argument. Area is continuous for convex Hausdorff convergence, so apply the argument to both A and $-A$. $\square$

This proof extends to fixed constant width in $\mathbb{R}^d$. It also applies to nonempty closed subclasses, for example an additional closed area constraint, or vanishing specified Fourier coefficients. If translation invariance is absent, confinement must instead be part of the problem. The condition “minimum width w ” alone gives no diameter bound; recall the extra sublevel argument in Session 8.

10.4 A regular hexagon around every constant-width body

Proposition 10.2 (Pál's covering lemma). *A planar body of constant width w admits a circumscribed regular hexagon whose opposite sides are distance w apart.*

Proof. Put $m(\theta) = h(\theta) - w/2$ and define

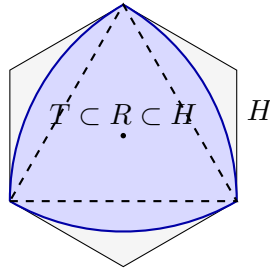
$$F(\theta) = m(\theta) - m(\theta + \pi/3) + m(\theta + 2\pi/3).$$

Since $m(\theta + \pi) = -m(\theta)$, $F(\theta + \pi/3) = -F(\theta)$. Continuity gives a zero θ_0 . Set $u_j = u(\theta_0 + j\pi/3)$, $j = 0, 1, 2$. Solve the two linear equations $c \cdot u_j = m(\theta_0 + j\pi/3)$ for $j = 0, 1$. The identity $u_0 - u_1 + u_2 = 0$ and $F(\theta_0) = 0$ give the third equation. Thus the three width- w supporting strips all have midlines through c . Their intersection is the required regular hexagon. $\square$

After translating c to zero, call the hexagon H . Its inradius is $w/2$, its side is $w/\sqrt{3}$, and

$$H = -H, \quad P(H) = 2\sqrt{3}w, \quad A(H) = \frac{\sqrt{3}}{2}w^2.$$

It has width w in its side-normal directions, but is not itself a constant-width body.



10.5 Chakerian's proof

Theorem 10.3 (Blaschke–Lebesgue). *Every planar convex body K of constant width w satisfies*

$$\boxed{\frac{\pi - \sqrt{3}}{2}w^2 \leq A(K) \leq \frac{\pi}{4}w^2.}$$

The lower equality holds exactly for Reuleaux triangles, up to rigid motions; the upper equality holds exactly for disks.

Proof of the bounds and attainment. The difference-body identity and the area polynomial yield

$$\pi w^2 = 2A(K) + 2V(K, -K).$$

Choose the centered hexagon above. Since $K, -K \subset H$, monotonicity gives

$$V(K, -K) \leq V(H, H) = A(H) = \frac{\sqrt{3}}{2}w^2.$$

Substitution proves the lower bound, attained by the Reuleaux triangle. For the upper bound use $P(K) = \pi w$ in $4\pi A(K) \leq P(K)^2$. The isoperimetric equality case gives the disk. $\square$

The short mixed-area argument proves the sharp lower bound without needing an unknown minimizer. It does not by itself establish lower equality rigidity: mixed area is not strictly increasing under inclusion. Session 11 supplies a geometric proof of that rigidity.

10.6 Why the hexagon estimate is sharp: a mixed-area plateau

Take the centered equilateral triangle T of side w , its Reuleaux triangle R , and $H = \text{conv}(R \cup (-R))$. Then $T \subset R \subset H$; H is the hexagon just used. The difference body $T + (-T)$ is a regular hexagon of side w , so

$$V(T, -T) = \frac{(3\sqrt{3}/2)w^2 - 2(\sqrt{3}/4)w^2}{2} = \frac{\sqrt{3}}{2}w^2 = V(H, -H).$$

Therefore *every* convex L with $T \subset L \subset H$ satisfies $V(L, -L) = \sqrt{3}w^2/2$ by monotonicity. Such an L need not have constant width. This calculation explains why replacing R by a much larger hexagon loses nothing in this particular mixed area.

Further mixed-volume perspective. For width one, minimizing $A(K)$ is equivalent to maximizing $V(K, -K)$. An optional identity in the instructor notes rewrites this as maximizing $P(S)/2 - A(S)$ for $S = \text{conv}(K \cup (-K))$. This is a reverse isoperimetric problem with geometric restrictions on S ; removing those restrictions changes the problem.

Sources. Chakerian, *Sets of constant width* (1966), Theorems 1–3; Yaglom–Boltyanskii, Chapter 7, Exercises 7-1, 7-7, 7-11, 7-12 and 7-15–16; manuscript SH0_2025_Curs9, pp. 5–13. For modern connections see [Bogosel, mixed volumes and Blaschke–Lebesgue](#), Sections 2–3, and [Antunes–Bogosel, support-function optimization](#), Sections 2–3. We use $V(K, L)$ for the mixed area denoted $A(K, L)$ in the first reference.

11 Geometric and variational proofs for constant width

By the end of the session: compare tangent-polygon approximations, understand a width-preserving perturbation, and connect the Blaschke–Lebesgue problem to finite-dimensional optimality conditions.

11.1 Equiangular tangent polygons

Choose $N \geq 4$ equally spaced unit normals u_j , where N is even, and let

$$Q_N(K) = \bigcap_{j=0}^{N-1} \{x : x \cdot u_j \leq h_K(u_j)\}.$$

These are circumscribed polygons with successive normal angles $\delta = 2\pi/N$. Zero-length sides are allowed when several supports meet at a corner. As $N \rightarrow \infty$, with the normals becoming dense, $Q_N(K) \rightarrow K$ in Hausdorff distance, and their areas and perimeters converge. A supporting polygon is generally not of constant width.

For fixed width w , its edge lengths satisfy

$$\ell_j = \frac{h_{j-1} + h_{j+1} - 2h_j \cos \delta}{\sin \delta}, \quad P(Q_N(K)) = 2 \tan(\delta/2) \sum_j h_j = Nw \tan(\pi/N).$$

Here $h_j = h_K(u_j)$, and $\sum_j h_j = Nw/2$ by pairing opposite directions. Passing to the limit gives another proof of Barbier’s theorem.

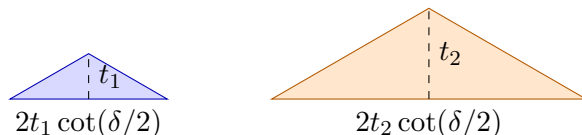
The two-cut calculation. In the construction from N to $2N$ sides, add the support in the middle of each normal interval. Two old adjacent supports and their opposites form a rhombus. The two new parallel supports remove isosceles triangles of depths $t_1, t_2 \geq 0$. Their sum and their total area are

$$s = t_1 + t_2 = w(\sec(\delta/2) - 1), \quad D = \cot(\delta/2)(t_1^2 + t_2^2). \quad (8)$$

Indeed, the opposite rhombus vertices are distance $w \sec(\delta/2)$ apart in this direction. A cut of depth t has base $2t \cot(\delta/2)$. Since

$$\frac{s^2}{2} \leq t_1^2 + t_2^2 \leq s^2,$$

equal cuts remove the least area; concentrating the cut at one end removes the most. Perimeter loss is linear in $t_1 + t_2$, so it is independent of the position of the new supporting strip.



Two removed corner triangles, translated for comparison. Both vertex angles are $\pi - \delta$; their areas are quadratic in depth.

11.2 A second proof of the bounds, including equality

Start with congruent regular circumscribed hexagons around a width- w disk D_w , a Reuleaux triangle R , and an arbitrary width- w body K . The hexagon around R has three alternating vertices on R . Refine their normal grids together: $N = 6, 12, 24, \dots$

At every refinement, the disk has $t_1 = t_2 = s/2$ in each pair. For R , one opposite support passes through an original corner, which stays a vertex of every supporting polygon. One cut depth is therefore zero. For K the cut loss lies between these two extremes. Sum over opposite pairs, whose cut triangles have disjoint interiors. Induction, starting from equal hexagon areas, yields

$$A(Q_N(R)) \leq A(Q_N(K)) \leq A(Q_N(D_w)).$$

Convex area continuity proves the two bounds of Session 10 again. This argument uses finite comparisons and a limit, so does not assume the existence of an unknown optimum.

Lower equality. The gap $A(Q_N(K)) - A(Q_N(R))$ never decreases during refinement. If its limit is zero, equality must hold already in each pair at the first refinement. A zero cut depth means that a new support passes through a hexagon vertex, and its contact point must be that vertex. The two old supports there constrain the interior corner angle of K to be at most 120° .

For any width- w body, a corner with exterior angle α produces an opposite circular arc of radius w and angle α : each support normal u at the corner p has opposite contact $p - wu$. Its endpoint distance is $2w \sin(\alpha/2) \leq w$, so $\alpha \leq \pi/3$. Thus the corner angle is at least 120° . Equality supplies an equilateral triangle of side w on the boundary. The corner–arc correspondence at its three vertices forces all three arcs of its Reuleaux triangle. Hence K is that Reuleaux triangle. The upper equality is already settled by isoperimetry; alternatively, equal paired cuts at every stage force central symmetry in the limit.

11.3 Reuleaux polygons and approximation

A *Reuleaux polygon* of width one is a constant-width body which is an intersection of finitely many unit disks. Arbitrary disk intersections need not have constant width. After removing redundant disks, the number of vertices and arcs is odd: $n = 2k + 1$. Label vertices counterclockwise $x_0, \dots, x_{n-1}$, modulo n . The arc centered at x_i joins x_{i+k} to x_{i+k+1} . If its central angle is θ_i , then

$$|x_i - x_{i+k}| = |x_i - x_{i+k+1}| = 1, \quad 0 < \theta_i \leq \pi/3, \quad \sum_i \theta_i = \pi.$$

Oddness follows by pairing each corner with its opposite arc in cyclic order; the paired features interlace around the boundary. The angle bound is the chord-diameter bound just proved; the sum is Barbier's formula, since each arc has radius one.

Proposition 11.1 (Reuleaux approximation). *Every planar body of constant width one is a Hausdorff limit of Reuleaux polygons of width one.*

Construction behind the statement. Take fine paired supporting lines, their contact polygon I , and their outer tangent polygon Q . For consecutive contacts a, b and their opposite contacts a', b' , let c be the intersection of the unit circles about a, b on the side of a', b' . Replace the arc ab by the unit arc about c ; replace the opposite arc by $a'c$ about a and cb' about b . The new opposite support contacts remain a vertex and an arc centered there, distance one apart. Repeat on the finitely many normal intervals. The resulting Reuleaux polygon lies between I and Q . Their Hausdorff gap tends to zero as the normal mesh tends to zero. The instructor notes discuss degeneracies and the gap estimate; Yaglom–Boltyanskii give the geometric construction on pp. 76–80.

11.4 Blaschke perturbations: preserving width while moving vertices

For a nondegenerate Reuleaux n -gon, $n \geq 5$, move x_0 slightly. Keep all other vertices fixed except x_k, x_{k+1} , which are determined by the continuous branches of the circle intersections

$$x_k(t) \in \partial B(x_0(t), 1) \cap \partial B(x_{n-1}, 1), \quad x_{k+1}(t) \in \partial B(x_0(t), 1) \cap \partial B(x_1, 1).$$

All required diameters remain one. For sufficiently small t , the cyclic order and positive arc angles persist, so the result remains a Reuleaux polygon. The two dependent vertices must be moved along with x_0 ; unconstrained movement of one disk center usually breaks width.

A *Blaschke perturbation* chooses $x_0(t)$ on the unit circle about x_k . Then x_k stays fixed and only two vertices move: x_0, x_{k+1} . Classical geometric proofs choose an area-decreasing such deformation and continue until vertices merge, reducing the number of arcs. Yaglom–Boltyanskii, Exercise 7-20, instead organize the reduction as a replacement of three arcs, starting with a shortest arc. Their comparison and its logic are detailed in the instructor notes.

11.5 The area gradient and a variational proof

The next formula is derived by integrating the normal displacement of the three moving arcs; it is not the gradient for independent disk centers. For $x_0(t) = x_0 + tv$ and the dependent vertices above, set

$$g_0 = \left(\tan \frac{\theta_k}{2} - \tan \frac{\theta_0}{2} \right) (x_0 - x_k) + \left(\tan \frac{\theta_{k+1}}{2} - \tan \frac{\theta_0}{2} \right) (x_0 - x_{k+1}). \quad (9)$$

Then $A'(0) = g_0 \cdot v$. At a local optimum every direction v is admissible for small positive and negative t , so $g_0 = 0$. The two diameter vectors are linearly independent. Hence $\theta_0 = \theta_k = \theta_{k+1}$. Apply this at every vertex to obtain equal arc angles π/n , and therefore a regular Reuleaux polygon. This reuses precisely the stationarity principle of Session 3.

For odd n , the area of the regular width-one example is

$$A_n = \frac{\pi}{2} - \frac{n}{2} \tan \frac{\pi}{2n}. \quad (10)$$

These values strictly increase with n and tend to $\pi/4$. They are derived from the skeleton polygon plus circular segments in the lab. In particular $A_3 = (\pi - \sqrt{3})/2$.

Closing the existence argument. In the class of Reuleaux polygons with *at most* N vertices, area attains its minimum. After translation all centers are bounded; take a convergent subsequence of their lists. The limit still has width one and is an intersection of at most N unit disks. Vertices may merge, which explains “at most”. If a minimizer has actual vertex count $n \geq 5$, stationarity makes it regular, but $A_n > A_3$. Thus the minimizer is a Reuleaux triangle. Approximation and area continuity pass the bound to all constant-width bodies. This final limit proves the inequality; equality rigidity comes from the tangent-polygon argument above, not from strict finite inequalities alone.

Optional connections. The instructor notes derive the gradient, give a Lagrange-multiplier formulation using the unit-diameter constraints, and show a negative second variation at regular examples with $n \geq 5$. The latter distinguishes a stationary configuration from a local minimum. The supplementary notebook checks the derivative and plots an admissible deformation of a Reuleaux pentagon.

Sources. Yaglom–Boltyanskii, Chapter 7, Exercises 7-10–12 and 7-20, with solutions pp. 247–253 and 260–264; [Bogosel, new variational arguments](#), Sections 2–4. The geometric proof is the core alternative to Chakerian; the full differential calculations are instructor material.